

Hypothesis Testing of Stock Returns

Week 3 — Statistical Inference in Finance

LEARNING OBJECTIVE

By the end of this lecture, you will be able to test whether a stock's average return is significantly different from zero (or from a benchmark) using a one-sample t-test.

KEY QUESTIONS

- Is this stock's return truly positive, or just lucky?
- How do we separate signal from noise in financial data?
- What does "statistically significant" actually mean for investors?

Setup: Null & Alternative

Formulating the hypothesis

NULL HYPOTHESIS (H_0)

$H_0: \mu = 0$ (the stock's mean return is zero)

ALTERNATIVE HYPOTHESIS (H_1)

$H_1: \mu \neq 0$ (the stock's mean return is non-zero)

WHY THIS MATTERS

- A positive average return could be due to random chance
- We quantify how surprising the data is, assuming H_0 is true
- If very surprising $\rightarrow$ reject $H_0 \rightarrow$ evidence of real return

Test Statistic

The one-sample t-statistic

FORMULA

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

WHERE

- $\bar{x}$ = sample mean return
- μ_0 = hypothesized mean (0 under H_0)
- s = sample standard deviation
- n = number of observations

INTUITION

Signal $(\bar{x} - \mu_0)$ divided by noise $(s / \sqrt{n})$. Larger $|t|$ = stronger evidence against H_0 .

Financial Example

Testing if AAPL monthly return > 0

DATA

- Stock: Apple Inc. (AAPL)
- Period: Jan 2020 – Dec 2023 (48 monthly returns)
- Sample mean ($\bar{x}$): 2.31% per month
- Sample std (s): 8.42% per month

CALCULATION

$$t = (2.31 - 0) / (8.42 / \sqrt{48}) = 1.901$$

Compare to t-distribution with $n-1 = 47$ degrees of freedom.

Python Implementation

Computing the test in code

```
import numpy as np
from scipy import stats

# Reproducible simulated AAPL monthly returns (%)
rng = np.random.default_rng(42)
returns = rng.normal(size=48)
returns = (returns - returns.mean()) / returns.std(ddof=1)
returns = returns * 8.42 + 2.31

# One-sample t-test (H0: mu = 0)
t_stat, p_value = stats.ttest_1samp(returns, 0)
print(f"t-statistic: {t_stat:.3f}")
print(f"p-value:      {p_value:.4f}")
print(f"mean return: {returns.mean():.2f}%")
```

Output:
t-statistic: 1.901
p-value: 0.0635
mean return: 2.31%

Interpretation

What does this result mean?

RESULT

- $t = 1.901$, two-tailed p-value = 0.0635
- At 5% significance level: $p > 0.05 \rightarrow$ fail to reject H_0
- At 10% significance level: $p < 0.10 \rightarrow$ reject H_0

FINANCIAL CONCLUSION

- Evidence of positive return is suggestive but not strong
- Typical in finance — returns are noisy, sample sizes limited
- Real-world: combine with other evidence (fundamentals, market regime)

COMMON MISTAKE

" $p = 0.06$ means there's a 6% chance H_0 is true" — WRONG. p is the probability of seeing this data *assuming* H_0 is true.

Apply this yourself

STUDENT EXERCISE (15 MINUTES)

Dataset: Monthly returns for Tesla (TSLA), Jan 2020 – Dec 2023

1. Compute the sample mean and standard deviation
2. Perform a one-sample t-test ($H_0: \mu = 0$)
3. Report the t-statistic and p-value
4. At $\alpha = 0.05$, do you reject H_0 ? What about $\alpha = 0.10$?
5. Write one sentence: what does this mean for an investor?